# K-EL/01 Technical QA Review

*Independent consistency audit against the supplied challenge brief*

## Executive finding

The package covers the assignment-stage deliverables and is more auditable than the earlier release. It is not proof of a validated actuator. Key design values are reproducible from the stated mass/COM inputs; CAD geometry is valid as a neutral concept model; component data are candidates and not a frozen system-level configuration.

| Disposition | Meaning |
|---|---|
| Covered for assignment | Core calculation, candidate component selection references, simplified arm/elbow STEP, labelled concept artwork, BOM, 3–5 page report, six critical-thinking answers, two risks, and 10-minute presentation prep. |
| Corrected in this revision | Removed legacy wording about a custom split-clamp sleeve/spindle where CAD uses a flexible coupling; updated encoder option to 512 CPT; explicitly converted N·m to N·mm for the shaft stress equation; recalculated load plane and overhang from CAD coordinates; corrected nominal bending stress to 7.23 MPa; added explicit forearm-link length to artwork and service-access caveat; interpreted EC-i speed/torque-gradient data using Maxon guidance on nonlinear iron-winding characteristics. |
| Still open by design | Vendor stack configuration, loaded efficiency/torque-speed/thermal point, dynamic/duty profile, brake/hold requirement, coupling hub selection, shaft and carrier detail, fits and life, detailed simulation and physical verification. |
| Not required by brief | Physical prototype, component purchasing, manufacturing-ready drawings and detailed FEA. No such work is claimed. |

## Source-to-submission coverage

| Original requirement | Evidence | QA outcome |
|---|---|---|
| Section 2: static torque | Calculation sheet and workbook; horizontal-arm calculation explicitly shown. | Pass for assignment calculation; does not cover dynamic inertia. |
| Section 2: speed/reduction | 45 RPM × 91 = 4095 RPM; motor nominal 4470 RPM cited. | Preliminary pass; only ~8.4% nominal speed headroom; loaded curve required. |
| Section 3A: torque target/margin | 12.1644 N·m using 2× factor. | Transparent assumption; not safety certification. |
| Section 3A: motor + gearbox evidence | Official manufacturer links and exact candidate numbers. | Shortlist only; configured combination not frozen. Ask vendor to clarify nominal Tω arithmetic vs 180 W class label; use actual EC-i speed/torque curve because its gradient is speed-dependent. |
| Section 3B: actuator + arm CAD | STEP neutral model, STEP builder source and FCMacro. | Envelope CAD, not detailed or manufacturing-ready parts; 21 solids in valid STEP. |
| Section 3B: assembly drawing/labelled view | One-page concept artwork plus numbered assembly drawing, each PDF/SVG/PNG. | Required view coverage is present; assembly/service access and bearing retention remain concept-level and must be checked in detailed CAD. |
| Section 3C: BOM and two risks | BOM.csv plus report risk section and risk register. | Covered at conceptual level; custom parts/materials/interfaces remain open. |
| Q1–Q6 critical thinking | Answers included in report and presenter Q&A. | Covered; design choices remain tentative. |
| Section 5 artwork | One-page drawing includes whole arm, enlarged exploded actuator, labels, load paths, dimensions, torque/speed callout and trade-off. | Covered for concept communication; image is not an assembly instruction or to-scale render. |
| Section 6: package organization | README, index, hashes, CSV registers, plan and technical presentation. | Covered; final archive must pass hash and zip integrity checks. |
| Section 8: final review | 10-minute presentation + 10-minute tech discussion prep. | Content supplied; actual live review is user-performed. |

## Numerical audit

| Quantity | Recalculation | Result / interpretation |
|---|---|---|
| Payload torque | 1.0 × 9.81 × 0.350 | 3.4335 N·m |
| Forearm/hand torque | 1.5 × 9.81 × 0.180 | 2.6487 N·m |
| Total / target | 6.0822 / 2 | 12.1644 N·m target |

| Quantity | Recalculation | Result / interpretation |
|---|---|---|
| Output power at target | 12.1644 × 45×2π/60 | 57.323 W ideal output |
| Motor speed | 45×91 | 4095 RPM; nominal margin 375 RPM |
| Eff. 40% screen | T/(91η); Pout/η; Tm/Kt | 0.3342 N·m; 143.3 W; ~3.64 A |
| Eff. 30% screen | Same equations | 0.4456 N·m; 191.1 W; ~4.85 A; fails simple continuous screen |
| Overhang moment | 24.525N × 50mm | 1.22625N·m = 1226.25N·mm |
| Stress | σb=7.228MPa; τ=35.852MPa; VM=62.517MPa | Nominal static only; corrected unit annotation |
| Backlash arc | 350mm×π/180 | 6.109mm at 1° average no-load |

## CAD review

- The exported STEP was re-imported through CadQuery and reported valid geometry; 21 solids were counted. This is a file/geometry sanity check, not a kinematic, interference, strength or assembly validation.
- Current model uses the ROTEX GS size 19 coupling envelope, independent Ø12 spindle, two 6201 bearing envelopes, split-yoke carrier, root plate, illustrative 300 mm upper-arm and 350 mm forearm envelopes.
- Bearing centres are Y=11 and 61 mm; forearm link centreline Y=111 mm; spacing and overhang are both 50 mm for the idealized reaction calculation. The model omits detailed seals, fit classes, screw threads, realistic clamp interfaces, cable routing, shoulder mount and detailed encoder installation.
- Native .FCStd was not generated because FreeCAD is not installed in the current build environment. The supplied .FCMacro is an execution route on a FreeCAD installation; STEP remains a CAD-equivalent neutral format allowed by the brief.

## Risk and validation verdict

- No full motor/gearhead operating point, configured order code or continuous thermal point is available in the files. Ask maxon for a written combination recommendation, rated output curve and explanation of the 180 W class label versus nominal Tω≈198 W arithmetic. Maxon support notes EC-i iron-winding speed/torque characteristics are nonlinear, so the listed 0.321 rpm/mNm gradient should not be extrapolated as a constant slope; it is not by itself evidence of a contradictory spec.
- No duty cycle/trajectory/acceleration/impact spec is defined. Get requirements, then calculate peak/RMS torque including inertia and worst-case stops; a static factor alone is insufficient.
- No brake/hold requirement is specified. If the robot must hold gravity after power loss, define a normally engaged brake or external mechanical support with a rated holding torque and safe state.
- Coupling hub bores, keying, service factor and torsional stiffness are not selected; do not treat size 19/98ShA as a full assembly part number.
- Shaft, bearing fits/life, axial load, spindle deflection, keyway-notch fatigue, flange fasteners and carrier bolts have only screening calculations or no calculations. Complete detailed drawings and analyses before manufacturing.
- Backlash may dominate endpoint accuracy: 1° is ~6.1 mm at payload radius before structural deflection and controls. An output-side encoder or lower-backlash reducer may be required depending on tolerance.
- Physical test, FEA, manufacturing detail and certification are outside this assignment's stated scope; label them as future validation, not deliverable failures.

## Final QA conclusion

The revised report/artwork/registries are internally aligned to one architecture. The deliverable set is assignment-complete as a concept package, with named closure actions for every material open engineering question. It should not be described as a complete engineered actuator release or proven safe design.